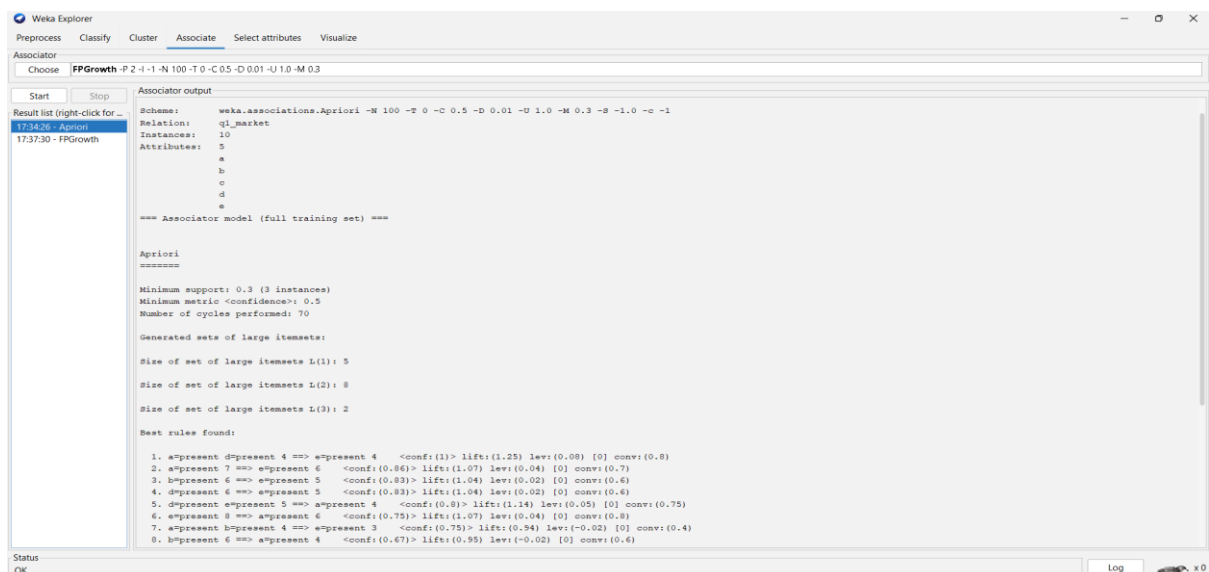


Q1. Consider the data set and perform the Apriori Algorithm and FP algorithm support:3 and confidence=50%.

CODE:

```
@relation q1_market
@attribute a {present}
@attribute b {present}
@attribute c {present}
@attribute d {present}
@attribute e {present}
@data
present,?,?,present,present
present,present,present,?,present
present,present,?,present,present
present,?,present,present,present
?,present,present,?,present
?,present,?,present,present
?,?,present,present,?
present,present,present,?,?
present,?,?,present,present
present,present,?,?,present
```

OUTPUT:



The screenshot shows the Weka Explorer interface with the 'Associate' tab selected. The 'FPGrowth' algorithm is chosen, and the 'Associator output' window displays the results of the Apriori algorithm. The output includes the minimum support (0.3), minimum confidence (0.5), and the number of cycles performed (70). It also lists the generated sets of large itemsets and the best rules found.

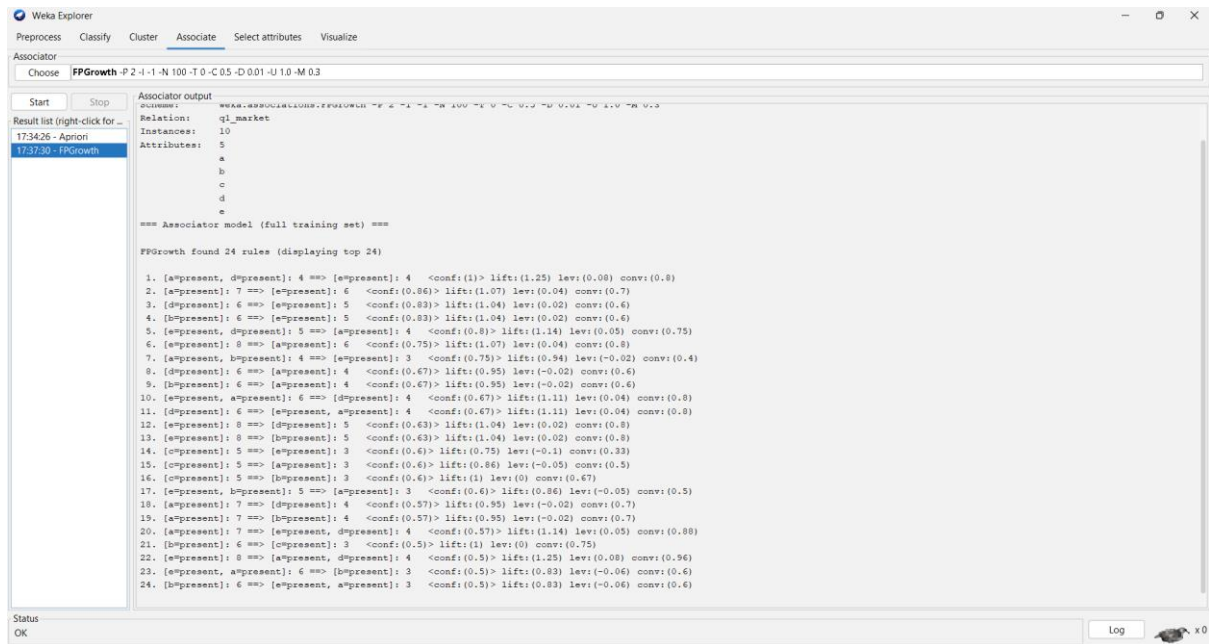
```
Result list (right-click for...):
17:37:30 - FPGrowth

Associator output
Start Stop
Choose FPGrowth -P 2 -I 1 -N 100 -T 0 -C 0.5 -D 0.01 -U 1.0 -M 0.3

Subsets: weka.associations.Apriori -N 100 -T 0 -C 0.5 -D 0.01 -U 1.0 -M 0.3 -S -1.0 -s -1
Relation: q1_market
Instances: 10
Attributes: 5
a
b
c
d
e

=== Associator model (full training set) ===

Apriori
=====
Minimum support: 0.3 (3 instances)
Minimum metric 'confidence': 0.5
Number of cycles performed: 70
Generated sets of large itemsets:
Size of set of large itemsets L(1): 5
Size of set of large itemsets L(2): 8
Size of set of large itemsets L(3): 2
Best rules found:
1. a=present d=present 4 ==> a=present 4 <conf:(1)> lift:(1.25) lev:(0.08) [0] conv:(0.8)
2. a=present 7 ==> a=present 6 <conf:(0.86)> lift:(1.07) lev:(0.04) [0] conv:(0.7)
3. b=present 6 ==> a=present 5 <conf:(0.83)> lift:(1.04) lev:(0.02) [0] conv:(0.6)
4. d=present 6 ==> a=present 5 <conf:(0.83)> lift:(1.04) lev:(0.02) [0] conv:(0.6)
5. d=present a=present 5 ==> a=present 4 <conf:(0.8)> lift:(1.14) lev:(0.05) [0] conv:(0.75)
6. a=present 8 ==> a=present 6 <conf:(0.75)> lift:(1.07) lev:(0.04) [0] conv:(0.8)
7. a=present b=present 4 ==> a=present 3 <conf:(0.75)> lift:(0.94) lev:(-0.02) [0] conv:(0.4)
8. b=present 6 ==> a=present 4 <conf:(0.67)> lift:(0.95) lev:(-0.02) [0] conv:(0.6)
```



Q2. Consider the market basket transactions shown in the above table.

(a) What is the maximum number of association rules that can be extracted from this data (including rules that have zero support)?

(b) What is the maximum size of frequent itemsets that can be extracted (assuming minsup > 0)?

CODE:

@relation q2_market

@attribute Milk {present,absent}

@attribute Beer {present,absent}

@attribute Diapers {present,absent}

@attribute Bread {present,absent}

@attribute Butter {present,absent}

@attribute Cookies {present,absent}

@data

present,present,present,absent,absent,absent

present,absent,absent,present,present,absent

present,absent,present,absent,absent,present

absent,absent,absent,present,present,present

absent,present,present,absent,absent,present

present,absent,present,present,present,absent

absent,absent,present,present,present,absent

absent,present,present,absent,absent,absent

present,absent,present,present,present,absent

absent,present,absent,absent,absent,present

OUTPUT:

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Associator

Choose Apriori -N 100 -T 0 -C 0.5 -D 0.01 -U 1.0 -M 0.3 -S -1.0 -c -1

Start Stop

Result list (right-click for...)

- 173426 - Apriori
- 173730 - FPGrowth
- 175126 - Apriori

Associator output

Scheme: weka.associations.Apriori -N 100 -T 0 -C 0.5 -D 0.01 -U 1.0 -M 0.3 -S -1.0 -c -1

Relation: q2_market

Instances: 10

Attributes: 6

- Milk
- Beer
- Diapers
- Bread
- Butter
- Cookies

=== Associator model (full training set) ===

Apriori

=====

Minimum support: 0.25 (3 instances)

Minimum metric (confidence): 0.5

Number of cycles performed: 65

Generated sets of large itemsets:

Size of set of large itemsets L(1): 12

Size of set of large itemsets L(2): 28

Size of set of large itemsets L(3): 26

Size of set of large itemsets L(4): 11

Size of set of large itemsets L(5): 2

Best rules found:

1. Bread=present 5 ==> Beer=absent 5 <conf:(1)> lift:(1.67) lev:(0.2) [2] conv:(2)
2. Butter=present 5 ==> Beer=absent 5 <conf:(1)> lift:(1.67) lev:(0.2) [2] conv:(2)
3. Butter=present 5 ==> Bread=present 5 <conf:(1)> lift:(2) lev:(0.25) [2] conv:(2.5)

Status

OK

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Associator

Choose Apriori -N 100 -T 0 -C 0.5 -D 0.01 -U 1.0 -M 0.3 -S -1.0 -c -1

Start Stop

Result list (right-click for...)

- 173426 - Apriori
- 173730 - FPGrowth
- 175126 - Apriori

Associator output

57. Milk=present Bread=present 3 ==> Beer=absent Butter=present 3 <conf:(1)> lift:(2) lev:(0.15) [1] conv:(1.5)

58. Milk=present Bread=present Cookies=absent 3 ==> Beer=absent 3 <conf:(1)> lift:(1.67) lev:(0.12) [1] conv:(1.2)

59. Milk=present Beer=absent Cookies=absent 3 ==> Bread=present 3 <conf:(1)> lift:(2) lev:(0.15) [1] conv:(1.5)

60. Milk=present Beer=absent Bread=present 3 ==> Cookies=absent 3 <conf:(1)> lift:(1.67) lev:(0.12) [1] conv:(1.2)

61. Milk=present Bread=present 3 ==> Beer=absent Cookies=absent 3 <conf:(1)> lift:(2.5) lev:(0.18) [1] conv:(1.8)

62. Milk=present Butter=present Cookies=absent 3 ==> Beer=absent 3 <conf:(1)> lift:(1.67) lev:(0.12) [1] conv:(1.2)

63. Milk=present Beer=absent Cookies=absent 3 ==> Butter=present 3 <conf:(1)> lift:(2) lev:(0.15) [1] conv:(1.5)

64. Milk=present Beer=absent Butter=present 3 ==> Cookies=absent 3 <conf:(1)> lift:(1.67) lev:(0.12) [1] conv:(1.2)

65. Milk=present Butter=present 3 ==> Beer=absent Cookies=absent 3 <conf:(1)> lift:(2.5) lev:(0.18) [1] conv:(1.8)

66. Milk=present Butter=present Cookies=absent 3 ==> Bread=present 3 <conf:(1)> lift:(2) lev:(0.15) [1] conv:(1.5)

67. Milk=present Bread=present Cookies=absent 3 ==> Butter=present 3 <conf:(1)> lift:(2) lev:(0.15) [1] conv:(1.5)

68. Milk=present Bread=present Butter=present 3 ==> Cookies=absent 3 <conf:(1)> lift:(1.67) lev:(0.12) [1] conv:(1.2)

69. Milk=present Butter=present 3 ==> Bread=present Cookies=absent 3 <conf:(1)> lift:(2.5) lev:(0.18) [1] conv:(1.8)

70. Milk=present Bread=present 3 ==> Butter=present Cookies=absent 3 <conf:(1)> lift:(2.5) lev:(0.18) [1] conv:(1.8)

71. Milk=absent Bread=absent Butter=absent 3 ==> Beer=present 3 <conf:(1)> lift:(2.5) lev:(0.18) [1] conv:(1.8)

72. Milk=absent Beer=present Butter=absent 3 ==> Bread=absent 3 <conf:(1)> lift:(2) lev:(0.15) [1] conv:(1.5)

73. Milk=absent Beer=present Bread=absent 3 ==> Butter=absent 3 <conf:(1)> lift:(2) lev:(0.15) [1] conv:(1.5)

74. Milk=absent Butter=absent 3 ==> Beer=present Bread=absent 3 <conf:(1)> lift:(2.5) lev:(0.18) [1] conv:(1.8)

75. Milk=absent Bread=absent 3 ==> Beer=present Butter=absent 3 <conf:(1)> lift:(2.5) lev:(0.18) [1] conv:(1.8)

76. Milk=absent Beer=present 3 ==> Bread=absent Butter=absent 3 <conf:(1)> lift:(2) lev:(0.15) [1] conv:(1.5)

77. Beer=present Diapers=present Butter=absent 3 ==> Bread=absent 3 <conf:(1)> lift:(2) lev:(0.15) [1] conv:(1.5)

78. Beer=present Diapers=present Bread=absent 3 ==> Butter=absent 3 <conf:(1)> lift:(2) lev:(0.15) [1] conv:(1.5)

80. Diapers=present Bread=present Butter=present 3 ==> Beer=absent 3 <conf:(1)> lift:(1.67) lev:(0.12) [1] conv:(1.2)

81. Beer=absent Diapers=present Butter=present 3 ==> Bread=present 3 <conf:(1)> lift:(2) lev:(0.15) [1] conv:(1.5)

82. Beer=absent Diapers=present Bread=present 3 ==> Butter=present 3 <conf:(1)> lift:(2) lev:(0.15) [1] conv:(1.5)

83. Diapers=present Butter=present 3 ==> Beer=absent Bread=present 3 <conf:(1)> lift:(2) lev:(0.15) [1] conv:(1.5)

84. Diapers=present Bread=present 3 ==> Beer=absent Butter=present 3 <conf:(1)> lift:(2) lev:(0.15) [1] conv:(1.5)

85. Diapers=present Bread=present Cookies=absent 3 ==> Beer=absent 3 <conf:(1)> lift:(1.67) lev:(0.12) [1] conv:(1.2)

86. Beer=absent Diapers=present Cookies=absent 3 ==> Bread=present 3 <conf:(1)> lift:(2) lev:(0.15) [1] conv:(1.5)

87. Beer=absent Diapers=present Bread=present 3 ==> Cookies=absent 3 <conf:(1)> lift:(1.67) lev:(0.12) [1] conv:(1.2)

88. Diapers=present Butter=present 3 ==> Beer=absent Cookies=absent 3 <conf:(1)> lift:(2.5) lev:(0.18) [1] conv:(1.8)

89. Diapers=present Butter=present Cookies=absent 3 ==> Beer=absent 3 <conf:(1)> lift:(1.67) lev:(0.12) [1] conv:(1.2)

90. Beer=absent Diapers=present Cookies=absent 3 ==> Butter=present 3 <conf:(1)> lift:(2) lev:(0.15) [1] conv:(1.5)

91. Beer=absent Diapers=present Butter=present 3 ==> Cookies=absent 3 <conf:(1)> lift:(1.67) lev:(0.12) [1] conv:(1.2)

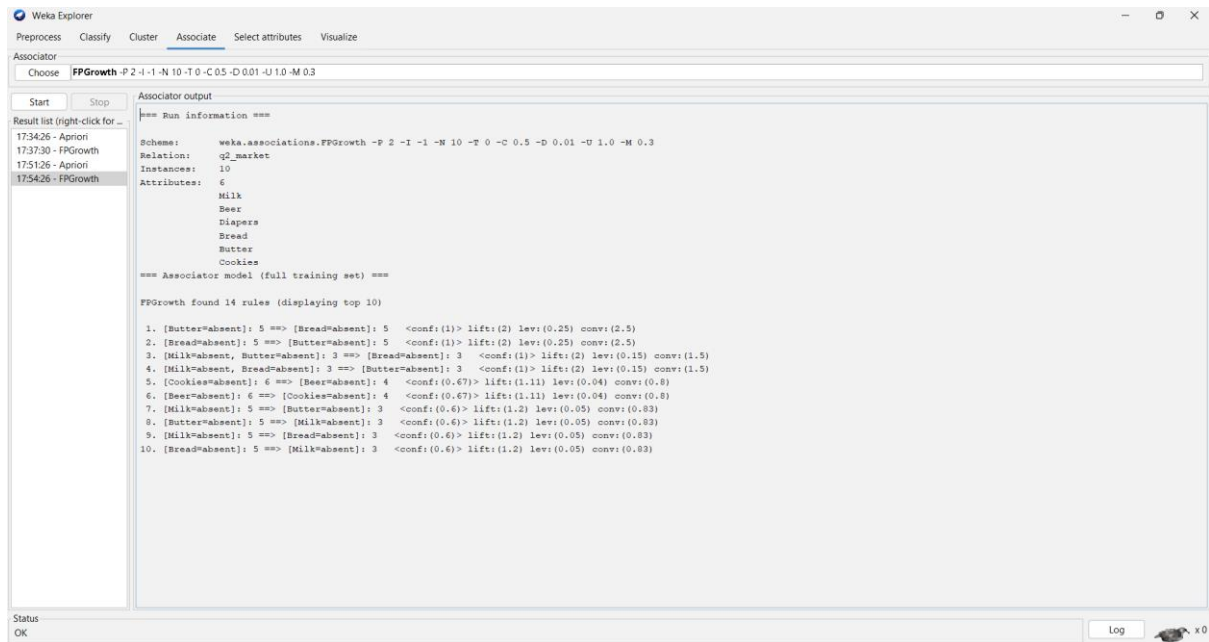
92. Diapers=present Butter=present 3 ==> Beer=absent Cookies=absent 3 <conf:(1)> lift:(2.5) lev:(0.18) [1] conv:(1.8)

93. Diapers=present Butter=present Cookies=absent 3 ==> Bread=present 3 <conf:(1)> lift:(2) lev:(0.15) [1] conv:(1.5)

94. Diapers=present Bread=present Cookies=absent 3 ==> Butter=present 3 <conf:(1)> lift:(2.5) lev:(0.18) [1] conv:(1.8)

Status

OK



Q3. Bayes classification and descion tree (using training and test data)

CODE:

@relation buys_computer

@attribute age {<=30,31...40,>40}

@attribute income {high,medium,low}

@attribute student {yes,no}

@attribute credit_rating {fair,excellent}

@attribute buys_computer {yes,no}

@data

<=30,high,no,fair,no

<=30,high,no,excellent,no

31...40,high,no,fair,yes

>40,medium,no,fair,yes

>40,low,yes,fair,yes

>40,low,yes,excellent,no

31...40,low,yes,excellent,yes

<=30,medium,no,fair,no

<=30,low,yes,fair,yes

>40,medium,yes,fair,yes

≤ 30 , medium, yes, excellent, yes

31...40, medium, no, excellent, yes

31...40, high, yes, fair, yes

> 40 , medium, no, excellent, no

OUTPUT:

The screenshot shows the Weka Explorer interface with the 'Classify' tab selected. The classifier chosen is 'J48 - C 0.25 - M 2'. The test options are set to 'Percentage split' at 70%. The classifier output is displayed in the main pane, showing the following details:

- Number of Leaves: 5
- Size of the tree: 8
- Time taken to build model: 0.01 seconds
- Time taken to test model on test split: 0 seconds
- Summary statistics:

Metric	Value	Percentage
Correctly Classified Instances	1	25 %
Incorrectly Classified Instances	3	75 %
Kappa statistic	-0.5	
Mean absolute error	0.75	
Root mean squared error	0.866	
Relative absolute error	150	%
Root relative squared error	164.3168	%
Total Number of Instances	4	

Detailed Accuracy By Class:

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	FPR Area	Class
yes	0.500	1.000	0.333	0.500	0.400	-0.577	0.250	0.417	yes
no	0.000	0.500	0.000	0.000	0.000	-0.577	0.250	0.500	no
Weighted Avg.	0.250	0.750	0.167	0.250	0.200	-0.577	0.250	0.458	

Confusion Matrix:

```
a b <-- classified as
1 1 | a = yes
2 0 | b = no
```

The screenshot shows the Weka Explorer interface with the 'Classify' tab selected. The classifier chosen is 'NaiveBayes'. The test options are set to 'Percentage split' at 70%. The classifier output is displayed in the main pane, showing the following details:

- Time taken to build model: 0 seconds
- Time taken to test model on test split: 0 seconds
- Summary statistics:

Metric	Value	Percentage
Correctly Classified Instances	2	50 %
Incorrectly Classified Instances	2	50 %
Kappa statistic	0	
Mean absolute error	0.4913	
Root mean squared error	0.5426	
Relative absolute error	98.2604	%
Root relative squared error	102.942	%
Total Number of Instances	4	

Detailed Accuracy By Class:

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	FPR Area	Class
yes	1.000	1.000	0.500	1.000	0.667	?	0.500	0.750	yes
no	0.000	0.000	?	0.000	?	?	0.500	0.583	no
Weighted Avg.	0.500	0.500	?	0.500	?	?	0.500	0.667	

Confusion Matrix:

```
a b <-- classified as
2 0 | a = yes
2 0 | b = no
```

Q4. Analysis the dataset “diabetes. csv” how the diabetes trend is for different age people, using linear regression and multiple regression.

CODE:

@relation pima_diabetes_numeric

@attribute 'preg' numeric

@attribute 'plas' numeric

@attribute 'pres' numeric

@attribute 'skin' numeric

@attribute 'insu' numeric

@attribute 'mass' numeric

@attribute 'pedi' numeric

@attribute 'age' numeric

@attribute 'class' numeric

@data

6,148,72,35,0,33.6,0.627,50,1

1,85,66,29,0,26.6,0.351,31,0

8,183,64,0,0,23.3,0.672,32,1

1,89,66,23,94,28.1,0.167,21,0

0,137,40,35,168,43.1,2.288,33,1

5,116,74,0,0,25.6,0.201,30,0

3,78,50,32,88,31.0,0.248,26,1

10,115,0,0,0,35.3,0.134,29,0

2,197,70,45,543,30.5,0.158,53,1

8,125,96,0,0,0.0,0.232,54,1

4,110,92,0,0,37.6,0.191,30,0

10,168,74,0,0,38.0,0.537,34,1

10,139,80,0,0,27.1,1.441,57,0

1,189,60,23,846,30.1,0.398,59,1

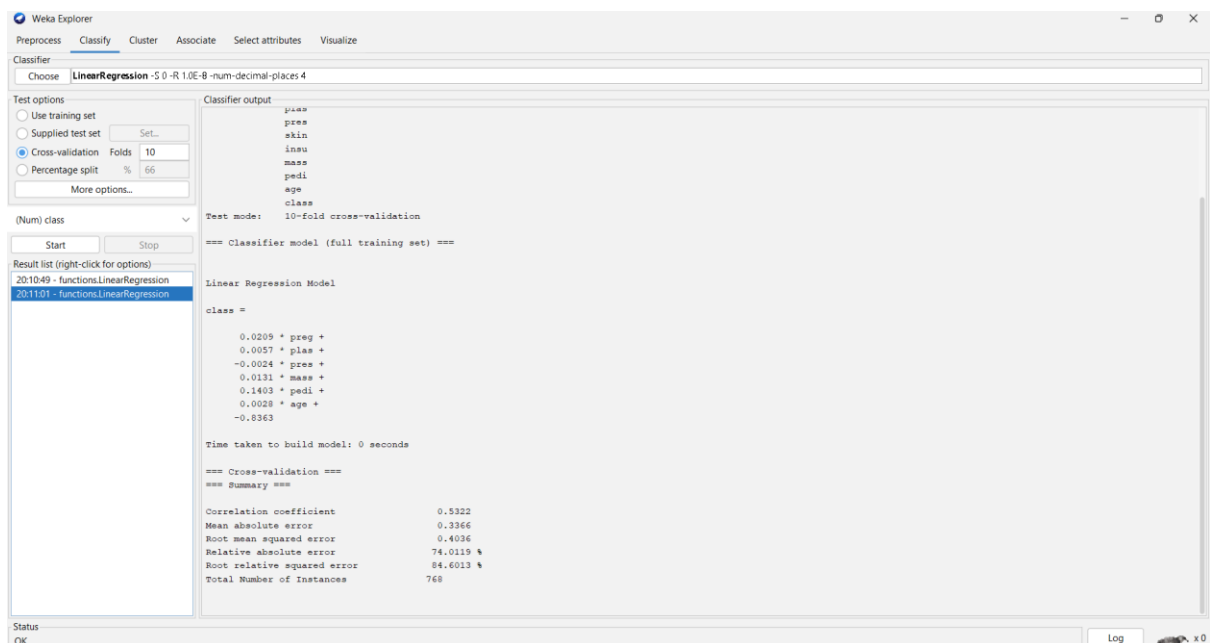
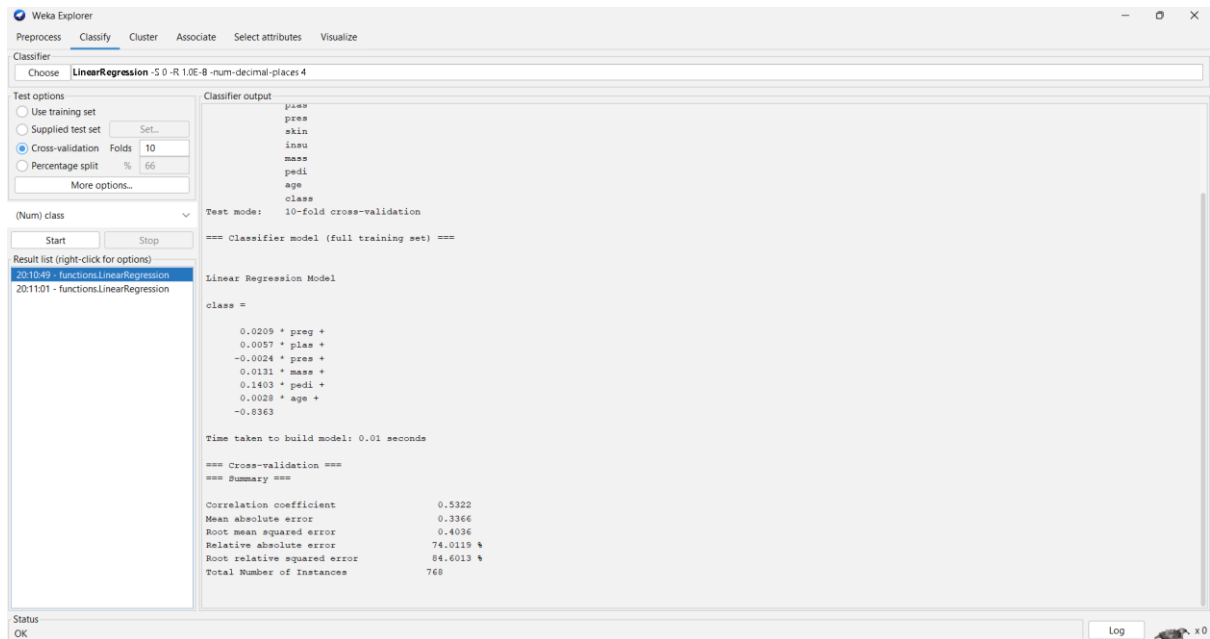
5,166,72,19,175,25.8,0.587,51,1

7,100,0,0,0,30.0,0.484,32,1

0,118,84,47,230,45.8,0.551,31,1

7,107,74,0,0,29.6,0.254,31,1

OUTPUT:



Q5. Implement using WEKA for the given Suppose a database has five transactions. Let min sup= 50%(2) and min con f= 80%.

Transactions

Items

T1

(M, O, N, K, E, Y)

T2

(D, O, N, K, E, Y)

T3

(M, A, K, E)

T4

(M, U, C, K, Y)

T5

(C,O, O, K, I,E)

- Find all frequent item sets using Apriori algorithm
- Also draw FP-Growth Tree

CODE:

@relation transactions

@attribute M {t, f}

@attribute O {t, f}

@attribute N {t, f}

@attribute K {t, f}

@attribute E {t, f}

@attribute Y {t, f}

@attribute D {t, f}

@attribute A {t, f}

@attribute U {t, f}

@attribute C {t, f}

@attribute I {t, f}

@data

t,t,t,t,t,f,f,f,f,f

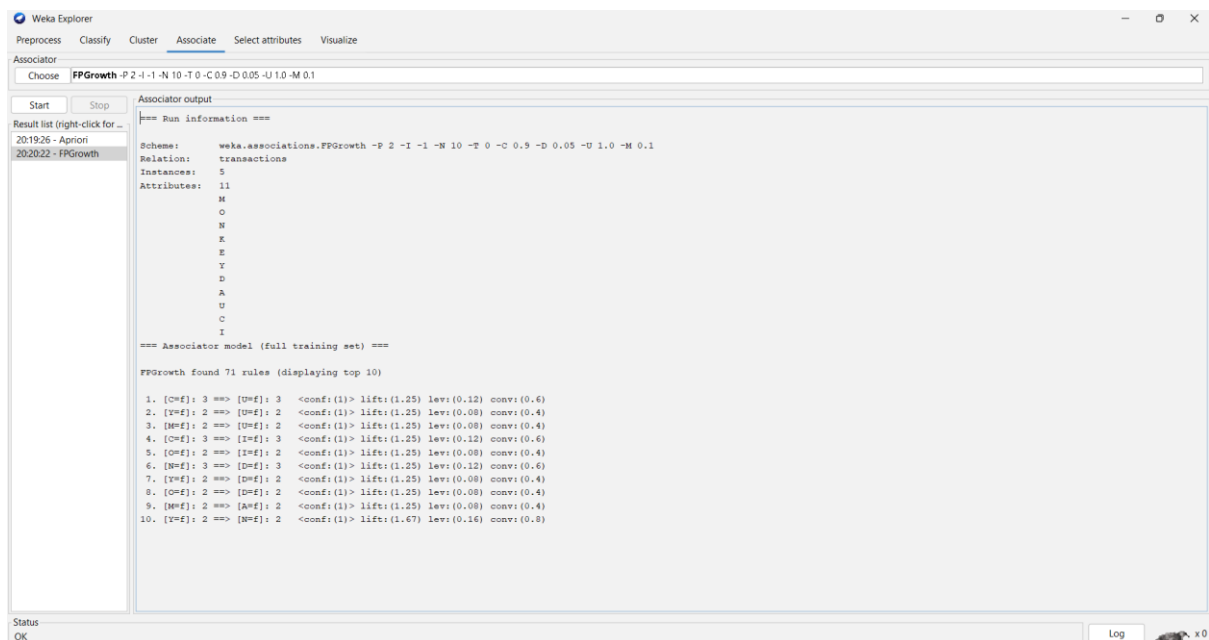
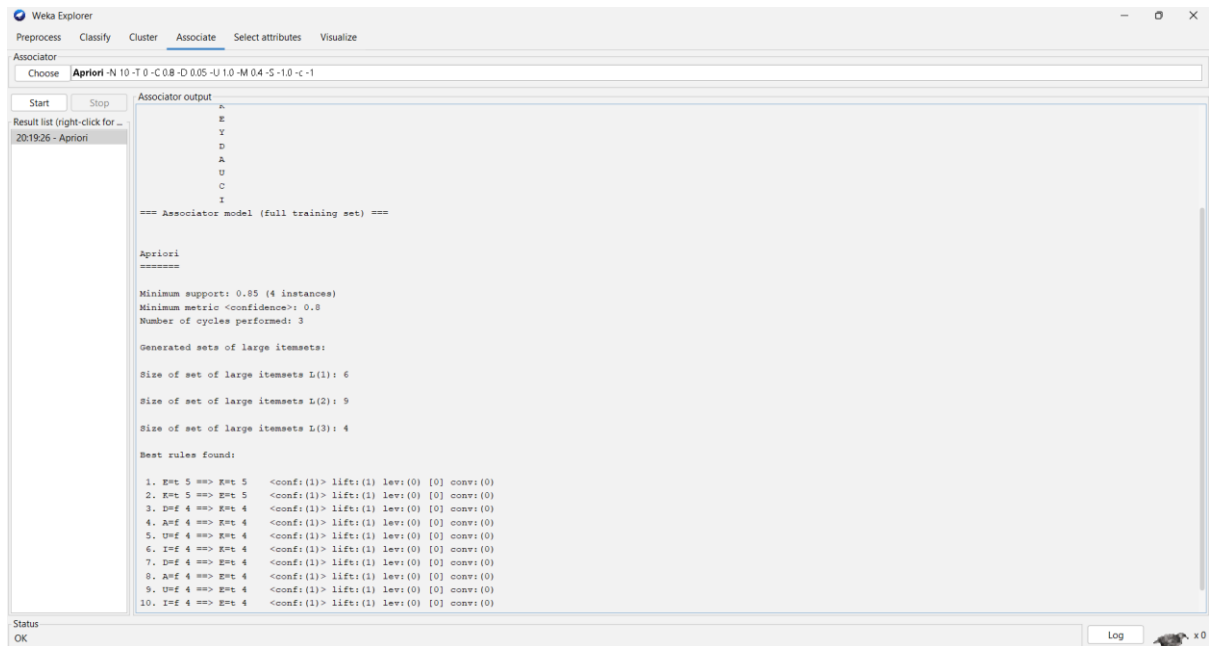
f,t,t,t,t,t,f,f,f,f

t,f,f,t,t,f,f,t,f,f,f

t,f,f,t,t,t,f,f,t,t,f

f,t,f,t,t,f,f,f,f,t,t

OUTPUT:



Q6. Prediction of Categorical Data using Decision Tree Algorithm through WEKA using any datasets. a) Tree b) Preprocess c) Logistic

CODE:

@relation weather

@attribute outlook {sunny, overcast, rainy}

@attribute temperature {hot, mild, cool}

@attribute humidity {high, normal}

@attribute windy {TRUE, FALSE}

@attribute play {yes, no}

@data

sunny,hot,high,FALSE,no

sunny,hot,high,TRUE,no

overcast,hot,high,FALSE,yes

rainy,mild,high,FALSE,yes

rainy,cool,normal,FALSE,yes

rainy,cool,normal,TRUE,no

OUTPUT:

The screenshot shows the Weka Explorer interface with the 'Classify' tab selected. The classifier chosen is 'J48 -C 0.25 -M 2'. The test options are set to 'Cross-validation' with 'Folds' set to '10'. The classifier output is displayed in the right pane, showing the run information and the full training set model.

```
=== Run information ===
Scheme:      weka.classifiers.trees.J48 -C 0.25 -M 2
Relation:    weather
Instances:    6
Attributes:   5
  outlook
  temperature
  humidity
  windy
  play
Test mode:    10-fold cross-validation

=== Classifier model (full training set) ===

J48 pruned tree
-----

windy = TRUE: no (2.0)
windy = FALSE: yes (4.0/1.0)

Number of Leaves    :    2
Size of the tree     :    3

Time taken to build model: 0.01 seconds
```

The screenshot shows the Weka Explorer interface with the 'Classify' tab selected. The classifier chosen is 'Logistic -R 1.0E-8 -M -1 -num-decimal-places 4'. The test options are set to 'Cross-validation' with 'Folds' set to '10'. The classifier output is displayed in the right pane, showing the run information and the full training set model.

```
=== Classifier model (full training set) ===

Logistic Regression with ridge parameter of 1.0E-8
Coefficients...

Variable          Class
=====
outlook=sunny      -17.9942
outlook=overcast   16.5885
outlook=rainy       6.779
temperature=hot     -6.779
temperature=mild    7.2424
temperature=cool    3.0990
humidity=normal     3.0990
windy=FALSE         37.5861
Intercept          -27.5777

Odds Ratios...

Variable          Class
=====
outlook=sunny      0
outlook=overcast   16006726.0253
outlook=rainy       879.1597
temperature=hot     0.0011
temperature=mild    1397.4411
temperature=cool    22.1944
humidity=normal     22.1944
windy=FALSE         2.1058434309323084E16

Time taken to build model: 0.02 seconds
```

Q7. Create the dataset using ARFF file format:

a. Find the frequent itemsets and generate association rules on this. Assume that minimum support threshold ($s = 33.33\%$) and minimum confidence threshold ($c = 60\%$).

b. List the various rule generated by apriori and FP tree algorithm ,mention wheather accepted or rejected.

CODE:

```
@relation market_basket
```

```
@attribute Hot_Dogs {t, f}
```

```
@attribute Buns {t, f}
```

```
@attribute Ketchup {t, f}
```

```
@attribute Coke {t, f}
```

```
@attribute Chips {t, f}
```

```
@data
```

```
t, t, t, f, f
```

```
t, t, f, f, f
```

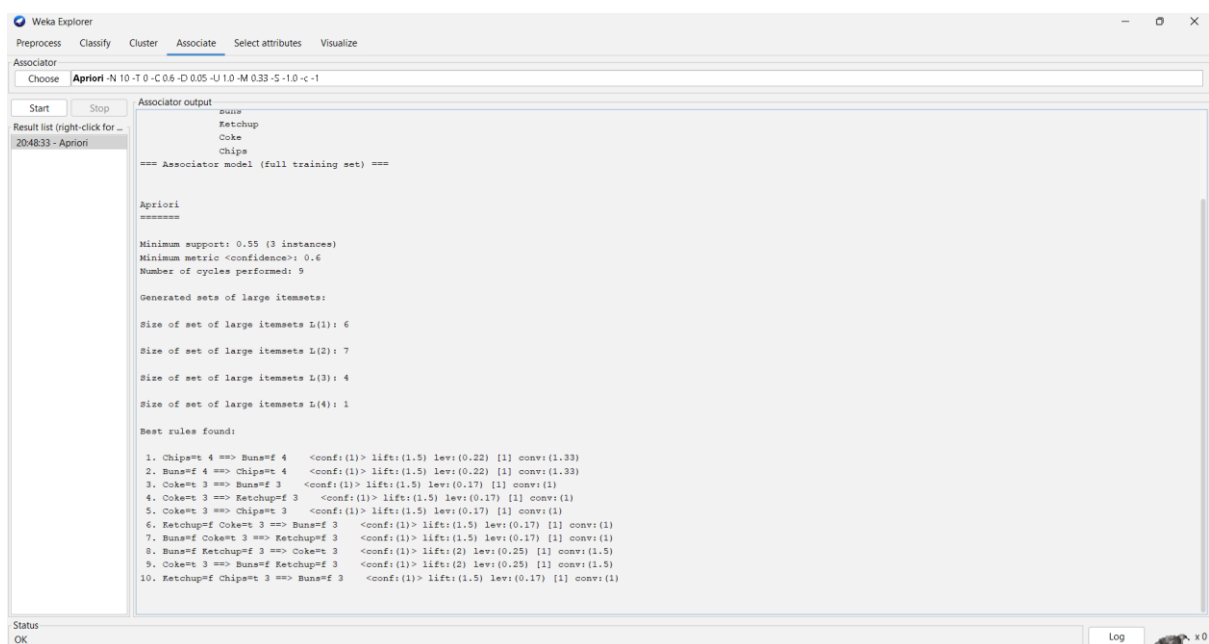
```
t, f, f, t, t
```

```
f, f, f, t, t
```

```
f, f, t, f, t
```

```
t, f, f, t, t
```

OUTPUT:



```
Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize

Associator:
Choose Apriori -N 10 -T 0 -C 0.6 -D 0.05 -U 1.0 -M 0.33 -S -1.0 -C -1

Start Stop
Result list (right-click for ...):
204833 - Apriori

Associator output:
Buns
Ketchup
Coke
Chips

=== Associator model (full training set) ===

Apriori
=====

Minimum support: 0.55 (3 instances)
Minimum metric <confidence>: 0.6
Number of cycles performed: 9

Generated sets of large itemsets:

Size of set of large itemsets L(1): 6
Size of set of large itemsets L(2): 7
Size of set of large itemsets L(3): 4
Size of set of large itemsets L(4): 1

Best rules found:
1. Chips=t 4 ==> Buns=f 4 <conf: (1)> lift: (1.5) lev: (0.22) [1] conv: (1.33)
2. Buns=f 4 ==> Chips=t 4 <conf: (1)> lift: (1.5) lev: (0.22) [1] conv: (1.33)
3. Coke=t 3 ==> Buns=f 3 <conf: (1)> lift: (1.5) lev: (0.17) [1] conv: (1)
4. Coke=t 3 ==> Ketchup=f 3 <conf: (1)> lift: (1.5) lev: (0.17) [1] conv: (1)
5. Coke=t 3 ==> Chips=t 3 <conf: (1)> lift: (1.5) lev: (0.17) [1] conv: (1)
6. Ketchup=f Coke=t 3 ==> Buns=f 3 <conf: (1)> lift: (1.5) lev: (0.17) [1] conv: (1)
7. Buns=f Coke=t 3 ==> Ketchup=f 3 <conf: (1)> lift: (1.5) lev: (0.17) [1] conv: (1)
8. Buns=f Ketchup=f 3 ==> Coke=t 3 <conf: (1)> lift: (2) lev: (0.25) [1] conv: (1.5)
9. Coke=t 3 ==> Buns=f Ketchup=f 3 <conf: (1)> lift: (2) lev: (0.25) [1] conv: (1.5)
10. Ketchup=f Chips=t 3 ==> Buns=f 3 <conf: (1)> lift: (1.5) lev: (0.17) [1] conv: (1)
```

Q8. Prediction of Categorical Data using Rule base classification and decision tree classification through WEKA using any datasets. Compare the accuracy using two algorithm and plot the graph.

CODE:

OUTPUT:

Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize

Classifier: Choose J48 -C 0.25 -M 2

Test options:
☐ Use training set
☐ Supplied test set
☒ Cross-validation Folds: 5
☐ Percentage split % 70
 More options...

(Nom) play
 Start Stop

Result list (right-click for options):
 203744 - trees.J48
 203929 - functions.Logistic
 210222 - trees.J48
 210222 - rules.J48

Classifier output

```

=== Predictions on test data ===

inst#  actual  predicted error prediction
1      1:yes   1:yes      0.000      1
2      2:no    1:yes      0.667      1
1      1:yes   2:no      0.667      1
1      1:yes   1:yes      0.000      1
1      2:no    1:yes      0.667      1
1      2:no    1:yes      0.000      1

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances      2      33.3333 %
Incorrectly Classified Instances    4      66.6667 %
Kappa statistic                    -0.3333
Mean absolute error                0.5000
Root mean squared error            0.6969
Relative absolute error            107.5362 %
Root relative squared error        125.2002 %
Total Number of Instances          6

=== Detailed Accuracy By Class ===

          TP Rate  FP Rate  Precision  Recall  F-Measure  MCC  ROC Area  PRC Area  Class
          0.667    1.000    0.400    0.667    0.500    -0.447  0.333    0.444    yes
          0.000    0.333    0.000    0.000    0.000    -0.447  0.333    0.500    no
Weighted Avg.  0.333    0.667    0.200    0.333    0.250    -0.447  0.333    0.472

=== Confusion Matrix ===
a b   <-- classified as
2 1 | a = yes
3 0 | b = no
  
```

Status: OK

Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize

Classifier: Choose JRip -F 3 -N 2.0 -O 2 -S 1

Test options:
☐ Use training set
☐ Supplied test set
☒ Cross-validation Folds: 5
☐ Percentage split % 70
 More options...

(Nom) play
 Start Stop

Result list (right-click for options):
 203744 - trees.J48
 203929 - functions.Logistic
 210222 - trees.J48
 210236 - rules.JRip

Classifier output

```

=== Predictions on test data ===

inst#  actual  predicted error prediction
1      1:yes   1:yes      0.5      1
2      2:no    1:yes      0.5      1
1      1:yes   2:no      0.667      1
1      1:yes   2:no      0.667      1
1      2:no    1:yes      0.667      1
1      2:no    1:yes      0.000      1

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances      1      16.6667 %
Incorrectly Classified Instances    5      83.3333 %
Kappa statistic                    -0.6667
Mean absolute error                0.6333
Root mean squared error            0.6557
Relative absolute error            119.6522 %
Root relative squared error        119.6522 %
Total Number of Instances          6

=== Detailed Accuracy By Class ===

          TP Rate  FP Rate  Precision  Recall  F-Measure  MCC  ROC Area  PRC Area  Class
          0.333    1.000    0.250    0.333    0.286    -0.707  0.056  0.417    yes
          0.000    0.667    0.000    0.000    0.000    -0.707  0.056  0.383    no
Weighted Avg.  0.167    0.833    0.125    0.167    0.143    -0.707  0.056  0.400

=== Confusion Matrix ===
a b   <-- classified as
1 2 | a = yes
3 0 | b = no
  
```

Status: OK